When Does Disconnection Have Empirical Content?

A Minimal Separability Framework, with Self-Refutation as a Consequence

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Abstract

The intuition that nothing is really connected to anything else—that each thing is a self-contained *orphan* and the felt connections are only its own reactions to its surroundings—is old and recurrent: Leibniz's windowless *monad*, Malebranche's *occasionalism*, and, re-equipped for the twentieth century, the operational closure of *autopoiesis* and the constructed worlds of radical constructivism. Its strong form asserts that no relation of any kind exists, with even the maximal container an orphan. I ask the only version of the question that can be wrong: *when does denying relations forbid an observation?* The answer is a **minimal separability framework**—the smallest set of epistemic commitments, fixed before the data, under which a disconnection claim has empirical content at all. There are three, and they are inseparable: a declared **decomposition** into intrinsic-property bearers, a declared **supervenience base** (Humean separability), and a **no-non-separable-whole counterfactual**. Pre-registration is their institutional implementation; Humean supervenience supplies the measurable content. The framework is claimed as **necessary only**: whether such commitments *suffice* inherits an open problem (they do not pin a unique decomposition). **Demarcation follows as a consequence**—the thesis is contentful only as a separability claim, and that claim collides with quantum non-separability (Bell)—and a second, sharper consequence is structural: the thesis's own positive mechanism, "connection is the thing's reaction to signals," **re-imports the relation it denies**, since a received signal is a relation (the self-refutation wall). The capstone—"even the totality is an orphan"—is shown to equivocate between *no external relatum* (trivially true) and *no internal relations* (the contested claim). A final, symmetric result is that, in their strong readings, **holism and disconnection are empirical twins**—each dissolves to a heuristic lens plus an unfalsifiable metaphysics, and neither forbids an observation the other permits. A worked demonstration is reported in the appendix.

1 Introduction

1.1 A thesis that keeps being rediscovered

The intuition is old and recurrent: a thing is not really joined to anything else; it sits alone, and what we read as its connection to the world is only *its own reaction* to the signals that reach it. Leibniz (1714) gave the intuition its sharpest image—the **monad**, of which he wrote that it has "no windows through which anything could come in or go out" (§7), so that what looks like interaction is only the internally generated unfolding of each substance. Malebranche (1674) pushed the same severance one step further with **occasionalism**: created things never truly act on one another; their apparent causal commerce is *occasion*, not cause. Hume then secured the epistemic root—in the perception of one event followed by another "we never can observe any tie between them" (Hume 1748). Three centuries later the same structure reappears, formally re-equipped, in the life sciences and cognitive science: the **operational closure** of autopoietic systems, which the environment can *perturb* but never *instruct* (Maturana and Varela 1980); the

constructed worlds of radical constructivism, in which what a knower takes to be features of its surroundings are largely its own making (Glasersfeld 1995); and the **interface theory of perception**, on which what we perceive is a species-specific dashboard that need not resemble the world it tracks (Hoffman et al. 2015).

This paper takes seriously a maximally ambitious version of the intuition, of the kind a systems-minded thinker is naturally led to:

> *No relation of any kind holds between one system, thing, or entity and another. Everything is an orphan connected to nothing; the only thing that produces the sensation of connection is the thing's own reaction to the environment in which it lives—that is, the universe, understood not as a celestial body but as the maximal container. The whole itself is an orphan; the connections are false, and are the thing's own response to the signals, information, and events around it.*

Stated this way the thesis is exhilarating and, I will argue, mostly false—but **instructively** false, and false in a way that exactly mirrors its opposite. The interesting question is never "is everything disconnected?" (to which the answer is either *trivially redescribable as yes* or *no*), but: **when does denying that two things are related say something that could be wrong?** That question has a precise, defensible answer, and locating it is the contribution of this paper.

1.2 Three readings, and the vacuity hazard named up front

The thesis admits three readings that must never be allowed to share credit.

- **Weak / lens reading.** What a thing does in response to its surroundings is shaped largely by its *own structure*, not dictated by the input; so the connections, meanings, and affordances it reads into its world are in substantial part its own construction. This is true—it is the content of operational closure and structural determinism (Maturana and Varela 1980)—and it predicts little on its own. It survives as a heuristic.
- **Strong / ontological reading.** *No relations exist*; the relata are real, the relations are not; each thing is intrinsically self-contained, and the appearance of connection is always a false addition to the world. This is the announced thesis. I will argue it is **coherent metaphysics, not science**, and that it does not survive its own statement (section 3).
- **Epistemic / methodological reading.** Explain each system's behaviour from its own state plus its boundary conditions, without positing irreducible relational entities. This is methodological individualism; it is defensible, and is how much of physics and engineering proceed—but it is *silent on ontology*: it says "do not posit," not "there is none."

The danger that haunts all three is **vacuity**, and here it takes a specific and historically documented form: the strong thesis is **empirically equivalent to its own negation**. Two billiard balls meet; one moves off. The relation-realist says a causal relation obtained. The disconnection theorist says ball A is an orphan that internally generated a reaction "appropriate" to the signal it received, and B likewise—the collision-relation is false, a coincidence of internal responses. *No observation distinguishes the two stories.* This is not a hypothetical risk; it is exactly the structure Malebranche (1674) and Leibniz (1714) engineered on purpose—occasionalism and the pre-established harmony were *designed* to be indistinguishable from ordinary interaction. For them it was a virtue; for a thesis that wears the vocabulary of "signals" and "information" it is a demarcation problem. A doctrine consistent with every possible

observation forbids nothing, and a framework that forbids nothing is a re-description, not a theory (Popper 1959). (That infalsifiability is not meaninglessness is granted in section 5; the charge here is *over-claiming*, not nonsense.)

1.3 The claim, and the method

The claim. Stripped of the universal and ontological overreach, a single defensible thesis survives, and it is constructive:

> The smallest set of epistemic commitments under which a disconnection claim has empirical content is three, and they are inseparable: a declared **decomposition** of the world into intrinsic-property bearers (the "orphans"); a declared **supervenience base**—the world's state supervenes on those intrinsic properties plus their spatiotemporal arrangement (Humean separability, Lewis 1986); and a **no-non-separable-whole counterfactual**—no admissible perturbation yields a whole carrying relational facts irreducible to the parts. Pre-registration is their institutional implementation. This framework is **necessary** for "nothing is related" to be falsifiable rather than notation-only. Whether it is also **sufficient**—whether fixing these commitments *completely* blocks the occasionalist re-description—is left open; the framework does not pin a unique decomposition, and the paper states this rather than claiming the converse. **Demarcation follows as a consequence**: contentful only as a separability claim, the thesis is then *false* where separability fails—quantum non-separability being the confirmed case.

Everything else in the paper either earns this necessary condition or tests it against the systems and the arguments that break it.

The method. The framework is built first, from primary sources read first-hand; its *consequences*—what it forbids, and where it self-destructs—are then tested against the hard cases that fix its boundary. Each refutation-boundary is reported as a constraint the framework must satisfy, not explained away; the surviving framework is stated as what it *forbids*; and where the evidence is deflationary, the verdict is deflationary. The ancestors that the thesis revives (Leibniz, Malebranche) are *reclassified*, not refuted: their systems are coherent, and the cost of their coherence is their empirical idleness.

Roadmap. Section 2 locates the one place a disconnection claim acquires testable content—separability—and states the minimal framework: the three commitments, their minimality, and the open sufficiency limit. Section 3 draws the consequence—the pincer (vacuous when safe, false when contentful), the contraposition class that breaks the contentful reading (quantum non-separability), and the self-refutation wall on which the strong reading founders. Section 4 supplies the conceptual anatomy the framework presupposes: the distinction between a relation and a reaction, the occasionalist re-description and its observational idleness, and the coordination gap the thesis inherits by keeping Leibniz's windows shut while removing his harmoniser. Section 5 turns the framework on its own maximal object, shows the capstone equivocal, and gives the verdict: the empirical-twin theorem, the surviving core, and an explicit statement of what the paper does not claim. A worked demonstration is in section A.

1.4 Contributions

This paper makes the following contributions.

- A **minimal separability framework** for the empirical content of disconnection claims: three epistemic commitments—a declared decomposition, a declared supervenience base,

and a no-non-separable-whole counterfactual—fixed before the data, with pre-registration as their institutional implementation. Claimed as *necessary only*.

- A **self-refutation result** for the strong reading: the thesis's own positive mechanism (connection as reaction-to-signal) presupposes the relation it abolishes, so the strong thesis cannot state its own explanans without contradiction.
- An **empirical-twin theorem**: the strong disconnection thesis and the strong holonic thesis are empirically equivalent—each reduces to a heuristic lens plus an unfalsifiable metaphysics, and neither forbids an observation the other permits.
- A **deflationary adjudication** of the thesis against the state of the art, sub-claim by sub-claim, naming each ancestor (Leibniz, Malebranche, Hume, autopoiesis, constructivism) and each live antithesis (ontic structural realism, relational quantum mechanics, dependent origination).
- A **surviving core**, stated as a prohibition: structural-determinist projectivism forbids the naïve instructionist / transmission reading, while explicitly *not* asserting that no relations exist.

What existed, and what this paper adds.

Ingredient	What existed	What this paper adds
Windowless substance	Leibniz: the monad, informal, God-coordinated	the separability conditions under which it is contentful
False causation	Malebranche: occasionalism, indistinguishable by design	the demarcation diagnosis: idle exactly because indistinguishable
Humean separability	Lewis: a supervenience thesis	a falsifiability gate, coupled to pre-registration
Non-separability	Healey: a physics fact	relocated as the contraposition class that breaks the thesis
Structural determinism	Maturana–Varela: closure, perturbation not instruction	the surviving core, stated as a prohibition—and the self-refutation it forces
Holism vs. disconnection	treated as rival metaphysics	shown to be empirical twins

2 The minimal separability framework

The weak reading of section 1.2 forbids nothing, and the strong reading was shown to be empirically equivalent to its own negation. So the constructive question is sharp: is there *any* way to cash "nothing is related" that forbids an observation? There is exactly one, and this section isolates it, states the commitments it requires, shows them minimal, and marks the point past which the thesis inherits an open problem rather than solving it.

2.1 Where a disconnection claim acquires content

"No relation holds between a and b" is, taken neat, untestable: every relation the realist asserts can be re-described as a coincidence of the two relata's internal reactions (section 4.2). The claim acquires testable content at one place only — when it is cashed as **separability**. The world is *separable* when its total state supervenes on the intrinsic physical properties of its smallest parts together with their spatiotemporal arrangement, with no irreducible relations left over (Lewis 1986). This is the precise, physical cash-value of "everything is an orphan": the whole is nothing but its parts, propertied intrinsically and laid out in space and time.

Separability is not a philosopher's curiosity; it is the assumption physics actually puts under test, and the literature that examines it is explicitly the literature on *holism and nonseparability* (Healey and Gomes 2022). That is already diagnostic: the only contentful form of "no relations" is a claim about whether wholes reduce to their parts — a claim whose *denial* is the very holism the thesis set out to reject. The disconnection thesis, made testable, becomes a separability thesis; and a separability thesis lives or dies on the same physics as its holist opposite.

2.2 The three commitments

For a disconnection claim to be falsifiable rather than notation-only, three commitments must be fixed *before* the data, and they are inseparable.

1. **A declared decomposition** into intrinsic-property bearers — the "orphans." Which things are the parts must be stipulated in advance. Without a declared partition, any inconvenient relation is absorbed post hoc by re-drawing the boundary or re-describing the relation as one part's internal reaction. (This is the exact analogue of the partition-presupposition that Putnam's triviality result exploits for computation (Putnam 1988): a decomposition found *after* the fact can rescue any claim.)

2. **A declared supervenience base.** The total state supervenes on the intrinsic properties of those parts plus their spatiotemporal arrangement (Humean separability, Lewis 1986). This is what gives "no relations" a global truth-condition: it says the world carries no fact over and above the parts-and-arrangement facts.

3. **A no-non-separable-whole counterfactual.** No admissible perturbation yields a joint state carrying a relational fact irreducible to the parts' intrinsic states. This is the risky commitment — the one that can meet a fact and lose.

Pre-registration is their institutional implementation. The decomposition and the supervenience base are epistemic commitments about the *form* a description must take; declaring them before measurement is what converts the disconnection thesis from an always-available re-description into a claim that can be wrong. The discipline is identical to the one a holonic claim needs — which is the first hint of the symmetry developed in section 5.2.

2.3 Minimality

Each commitment does independent work, and none is redundant.

Each is required. Drop (1) and there are no declared relata, so "no relation between a and b" has no a and b to be about; the claim is vacuous. Drop (2) and "no relations" places no constraint on the global state — a non-separable world could satisfy it locally while the whole still carries irreducible facts; the claim forbids nothing. Drop (3) and what remains is a static picture, not a prediction: it describes a way of decomposing without risking anything against a measurement.

No pair suffices. Commitments (1)+(2) without (3) give a metaphysical picture with no pre-registered risk — separability asserted, never exposed. Commitments (1)+(3) without (2) leave "irreducible relational fact" undefined, since without a supervenience base there is no saying what a whole would have to reduce *to*. Commitments (2)+(3) without (1) have no bearers to carry the intrinsic properties the base quantifies over.

No fourth does logical work. Candidate additions — that the parts be spatially separated, that causation be local, that influences propagate continuously — are either consequences of (2)

and (3) or independent physical hypotheses that the disconnection thesis does not need in order to have content. The framework is three commitments, not two and not four.

2.4 Necessary only; the open sufficiency limit

The framework is claimed as **necessary**: without these three commitments, fixed in advance, a disconnection claim has no empirical content. It is *not* claimed sufficient, and the reason is exact. The commitments do not pin a *unique* decomposition. Different admissible partitions can disagree on whether a given whole is separable — separability, like near-decomposability, is a property of a *(system, decomposition, signature)* tuple, not of the world simpliciter. So fixing the three commitments narrows, but does not close, the occasionalist re-description: one can still ask whether *some* admissible decomposition always restores separability, and that question inherits the same underdetermination occasionalism was built to exploit.

This is the disconnection thesis's version of the open problem the holonic thesis inherits from Putnam and Chalmers. There the question was whether pre-registered partition-admissibility *completely* blocks the triviality re-description; here it is whether it completely blocks the occasionalist re-description. The paper states the limit and does not claim the converse. What follows — the consequences, the walls, the verdict — rests only on the *necessity* claim, which is all the argument needs.

3 The consequence: the pincer, the contraposition class, and the self-refutation wall

The framework is only as good as the consequences it forces. This section draws them. Two are fatal to the strong reading and one is methodological; together they pin the thesis between a wall of vacuity and a wall of fact, and then knock out the strong reading from inside.

3.1 The pincer

The disconnection thesis is caught between two failures, with no position between them.

Vacuous when safe. In its uncashed form — "the relations are false, only the orphan's reactions are real" — the thesis is empirically equivalent to relation-realism (section 4.2). It forbids no observation and so, as a claim about the world, says nothing testable (Popper 1959).

False when contentful. Cashed as separability (section 2.1), the thesis at last forbids something — non-separable wholes — and is thereby exposed. The exposure is not hypothetical: the forbidden thing exists (section 3.2).

There is no third setting of the dial. Every increment of content the thesis buys, it buys by approaching separability, and separability is where the physics is waiting. The thesis is contentful only on the approach to its own refutation.

3.2 The contraposition class: quantum non-separability

On the adversarial-demarcation method, the systems a thesis calls impossible are its empirical commitment. The contentful disconnection thesis calls non-separable wholes impossible. They are real.

Entangled states do not factor into subsystem states: the joint state carries correlations that no assignment of intrinsic properties to the parts, plus their arrangement, can reproduce.

Bell (1964) showed that any locally-separable (local-hidden-variable) model obeys an inequality that quantum mechanics violates; the violation has been confirmed experimentally to high significance (Aspect et al. 1982). The standard reading of this body of work is precisely a failure of separability: the whole possesses relational facts irreducible to the intrinsic states of its parts (Esfeld 2004; Healey and Gomes 2022). Commitment (3)'s counterfactual is therefore violated by nature, and the contentful disconnection thesis is *false* wherever entanglement obtains.

Two honest caveats, neither of which rescues the thesis. *First*, non-separability admits interpretation-relative descriptions — a pragmatist account can reproduce the correlations without asserting metaphysical holism (Healey and Gomes 2022) — but every interpretation that keeps both locality and a single outcome concedes the non-separability; the thesis cannot help itself to the exception without naming which interpretation it is buying. *Second*, the formal escapes that *would* preserve separability — superdeterminism, certain retrocausal or hidden-variable manoeuvres — purchase it by pre-coordinating the parts' states in advance, which is structurally the occasionalist / pre-established-harmony move (section 4.4): they restore separability only by making it observationally idle again. The rescue and the vacuity are the same move. (These escapes are, in the verification ledger of section 5.6, the weakest sources; nothing in the argument promotes them to fact.)

The collision has a clean dual. Ontic structural realism reads exactly this physics the opposite way — relations as fundamental, relata as derivative (Ladyman and Ross 2007) — and relational quantum mechanics makes a system's very properties hold only relative to other systems (Laudisa and Rovelli 2024; Rovelli 1996). The disconnection thesis (relata without relations) and ontic structural realism (relations without relata) are inversions of one another; entanglement is the pressure point where they meet, and for the *contentful* question it favours the relational side.

3.3 The self-refutation wall

The deepest objection is not the physics; it is that the strong thesis cannot state its own mechanism. Its positive proposal is that the appearance of connection is "the thing's reaction to the signals, information, and events around it." But a received signal *is* a relation: a channel from a source to a receiver, whose information content is, by construction, a *mutual* and therefore symmetric relation between the two. "Reaction *to* the environment in which it *lives*" is doubly relational already in its grammar. And the thesis's own best ally says so outright: autopoietic systems are operationally closed but **structurally coupled** to their medium (Maturana and Varela 1980) — and structural coupling is a relation. The thesis recruits structural determinism and discards the one thing structural determinism insists on.

So the strong reading must assert, in the same breath, "no relations exist" (its predicate) and "the appearance of connection is the system's relation-mediated reaction to signals from its surroundings" (its explanans). The load-bearing joint is a contradiction: the thesis denies relations in what it claims and presupposes them in how it explains. It refutes itself the moment it explains anything.

The only escape is to deny that any signal is really received — to say a hidden coordinator produces the matched internal reactions (occasionalism). That escape (i) reintroduces a coordinator the thesis was meant to do without, and (ii) collapses the thesis back into the idle re-description of section 3.1. There is no version of the strong thesis that both explains the appearance of connection and denies all relations. This is the same self-administered *modus tollens* that fells the maximal claim in section 5.1 — here applied not to the object but to the mechanism.

3.4 What the bounded thesis forbids

Strip the strong reading away and the surviving *contentful* claim — separability under a declared decomposition — still forbids something pre-registrable and risky:

> A whole whose joint statistics, under the declared decomposition, cannot be reproduced by any assignment of intrinsic properties to its parts plus their arrangement is **not** an aggregate of orphans.

This is testable and, in the quantum regime, already failed: Bell-violating joint statistics are exactly such a non-reproducible signature. The forbidden class is non-empty, which is what makes the bounded thesis more than a perspective — and what makes the strong thesis false. The surrendered territory is large: plausibly the entire quantum foundation, and with it any claim that the world's fundamental level is an aggregate of orphans. That is the price of "no relations."

4 Conceptual anatomy: relation, reaction, and the idle re-description

The framework and its consequences presuppose a small anatomy of the thesis: the conflation it lives on, the formal shape of the re-description that makes it idle, the ladder of distinct things "disconnection" can mean, and the coordination debt it inherits from its own ancestry. This section supplies them.

4.1 Relation versus reaction

The thesis trades on a single equivocation: *reaction* sounds intrinsic, *relation* sounds external, and the rhetoric of "the orphan merely reacts" borrows the first to deny the second. But the equivocation does not survive disambiguation. "X reacts to a signal" has two readings. On the *intrinsic-change* reading, X changes by its own laws once triggered — true, and untouched by anyone; but the trigger is a relation, so this reading does not abolish relations, it only insists the response is the responder's own. On the *no-relatum* reading, there is no source at all — which the thesis cannot assert, because its own statement gives the thing "the environment in which it lives."

What is left, once the equivocation is removed, is the defensible residue: the *response is structure-dependent, not input-dictated.* That is a claim about the *form* of the relation between system and environment — that the system contributes most of the shape — not a claim that the relation does not exist. It is constructivism, and it is true (section 5.3); it is not relational nihilism.

4.2 The occasionalist re-description and its idleness

Why is the uncashed thesis empirically equivalent to its negation? Because the re-description is total and lossless. Take any relational model in which a fact $R(a, b)$ holds between two things. Replace it with two intrinsic functions, $f_a(\sigma)$ and $f_b(\sigma)$, each mapping the "signal" σ to an internal reaction, plus a coordination assumption that the two reactions agree wherever R said they would. The rewrite reproduces every observable consequence of R and adds nothing measurable — it merely relocates the relational fact into a coordination assumption held off-stage. This is not a weakness the thesis can patch; it is the design. Malebranche (1674) made created

causation an *occasion* rather than a cause precisely so that the world would look identical either way; Leibniz (1714) had the monads adapted to one another in advance by God (§§51–52) and held the relation between two things to be "a merely ideal thing" added by the mind, not a constituent of either (5th letter to Clarke, §47); the Stanford surveys of both doctrines document this indistinguishability (Bobro 2005; Lee 2020).

Idleness is not falsehood. The occasionalist picture is coherent metaphysics — it just forfeits any claim to be a *discovery about signals*, because no signal-level observation could favour it over the relational picture it mimics. The thesis may keep the metaphysics or keep the empirical pretension; it cannot keep both.

4.3 The three things "disconnection" can mean

As with the three readings of section 1.2, three distinct claims hide under one word, and they must never share credit. Ranked by content:

1. **No external relatum for the totality.** The maximal container has nothing outside it to be related to. Trivially true, and empty (section 5.1).
2. **Response is structure-determined, not input-dictated.** True, with modest content: it forbids the naïve instructionist reading on which the environment stamps its structure into a passive receiver (section 5.3).
3. **No relations exist at all.** Relational eliminativism: maximal content, and both self-refuting (section 3.3) and false (section 3.2).

The thesis draws its excitement from (iii), its defensibility from (ii), and its air of obviousness from (i), sliding among them as challenged. Naming the three and refusing the slide is most of the adjudication.

4.4 The godless-monadology coordination gap

The sharpest way to see the strong thesis fail is to notice what it inherited and what it threw away. It is a **secularised, environment-driven monadology**: it keeps Leibniz's windowlessness and his false-relations, but deletes the *pre-established harmony* — and the harmony was the whole point. Leibniz introduced a God who synchronised the windowless monads precisely to answer the coordination problem: if nothing acts on anything, why do the substances' internal states agree, so that the world hangs together as one coherent scene? (Bobro 2005; Leibniz 1714) Delete the harmoniser and put "reaction to a real environment" in its place, and the thesis faces a clean dilemma:

- either the environment *really acts* on the thing — in which case a relation obtains, and the thesis self-refutes (section 3.3);
- or it does not, and the matched reactions are pre-coordinated — in which case the thesis needs Leibniz's God, or an equivalent off-stage coordinator, and is idle (section 4.2).

There is no third branch. The thesis inherits Leibniz's hardest question and discards his answer; its modern vocabulary of "signals" hides the debt without paying it.

It is worth marking the opposite pole, because it shows the disconnection thesis is not the only way to deny intrinsic isolation's comforts. The most thoroughgoing anti-substantialist view on offer — the Madhyamaka doctrine of dependent origination — reaches the *reverse*

conclusion from a neighbouring starting point: nothing has intrinsic, independent existence *because* everything is dependently originated, related all the way down (Garfield 1995). Where the disconnection thesis wants relata without relations, dependent origination dissolves the relata *into* the relations. Both reject the naïvely self-standing object; only one of them can also keep saying the connections are false.

5 The maximal object, and the verdict

The framework has one object left to face: its own maximal one. The thesis ends by declaring the totality an orphan; turned on that declaration, the framework shows it ill-formed, and the diagnosis opens onto the paper's final, symmetric result.

5.1 The capstone: "even the totality is an orphan"

"The whole itself is an orphan" equivocates between two claims the framework has already separated. On one reading it says the totality has *no external relatum* — nothing outside the maximal container to relate to. That is trivially true and empty: it is a fact about the word "maximal," not a discovery about connection. On the other reading it says the totality has *no internal relations* — that the cosmos is a mere aggregate of intrinsically self-standing parts. That is the contested strong claim, and it is exactly what entanglement (section 3.2) and the self-refutation wall (section 3.3) deny.

The capstone trades the first for the second under cover of the single word "orphan," just as the strong reading traded "reaction" for "no relation." And the maximal object is where the trade is most visible, because the strongest live position about the cosmos-as-whole runs the other way: priority monism takes the cosmos to be the *one* fundamental whole, with its parts derivative (Schaffer 2010). Monism and the disconnection thesis agree that the cosmos is the maximal object and draw opposite morals from it — whole-before-parts versus parts-without-whole — which is precisely the ambiguity the capstone needs no one to notice.

5.2 Holism and disconnection as empirical twins

Step back and the symmetry is exact. Consider the maximally strong *holonic* thesis — every system is constitutively related, the universe the maximal holon — and the maximally strong *disconnection* thesis — no system is related, the universe the maximal orphan. In their strong, ontological readings these are **empirical twins**:

- each is consistent with every possible observation — the holonic thesis because, under unrestricted composition, everything is a part of some whole; the disconnection thesis because, under occasionalist re-description, every relation is re-readable as coincident internal reactions;
- neither forbids an observation the other permits;
- each survives only in two registers — a heuristic *lens* that organises description, and an *unfalsifiable metaphysics* that asserts more than any measurement could check.

The corollary is the result. *A doctrine and its negation that make the same predictions are both lenses.* At most one of two contradictories can be an ontological discovery, and when the two are empirically indistinguishable, the parsimonious reading is that neither is — that the "everything is related / nothing is related" axis is a choice of descriptive lens projected

onto the world, not a fact read off it. The contentful residues differ in name — the holonic thesis cashes out as near-decomposability, the disconnection thesis as separability — but both reduce to a *(system, decomposition, signature)* tuple, and both collide with the same physics: non-separability refutes "every system is near-decomposable" and "every whole is separable" alike. The mirror is not a coincidence; it is what one should expect of a thesis and its negation once each is made testable.

5.3 The surviving core

What survives the two walls is not a metaphysics of disconnection but a methodological prohibition, and it is worth stating in its own right:

> A system's response to its surroundings is shaped — often dominantly — by the system's own structure, not dictated by the input; so the connections, meanings, and affordances it reads into its world are in substantial part its own construction, and must not be read naïvely back as intrinsic features of the environment merely transmitted into a passive receiver.

This is the content of operational closure and structural determinism (Maturana and Varela 1980), of radical constructivism (Glasersfeld 1995), and of the eigenform tradition on which an object is a token for a stable behaviour of the observer rather than a readout of a thing-in-itself (Foerster 1981; Kauffman 2003). What it *forbids* is the naïve instructionist / transmission reading, and it is weakly testable: the same perturbation drives structure-dependent, not input-dictated, responses. What it does *not* say is that no relations exist (it presupposes the system–environment relation), that connections are "false" (constructed and system-relative is not nonexistent), or anything at all about the totality. It is a lens worth keeping — precisely the weak reading of section 1.2, retained as a discipline.

5.4 Verdict

The verdict is deflationary, and decisively so. The strong reading — the only one that asserts what relation-realism denies — falls to two independent fatal walls: it cannot state its own mechanism without the relations it abolishes (section 3.3), and in its one contentful form it collides with quantum non-separability (section 3.2). Between them it is held in the pincer of section 3.1: vacuous when safe, false when contentful. Its empirical equivalence with its own negation (section 4.2) means that, as a purported discovery about the world, it forbids no observation and is empirically vacuous.

The ancestors are *reclassified, not refuted*. Leibniz's windowless monads and Malebranche's occasionalism are coherent metaphysics; the cost of their coherence is the empirical idleness the thesis inherits the moment it adopts their indistinguishability. The thesis is best filed as a heuristic lens (structure-dependent response) wrapped around an unfalsifiable metaphysics of the Leibniz–Malebranche family — the same drawer in which the honest form of its holonic twin belongs.

5.5 What this paper does not claim

- **Not that the thesis is meaningless.** Unfalsifiable metaphysics is legitimate metaphysics, and the surviving constructivist core is genuinely useful. The charge is *over-claiming* — wearing the vocabulary of signals and information — not nonsense.

- **Not that relations are metaphysically fundamental.** The paper does not assume ontic structural realism or relational quantum mechanics. It claims only that the *contentful* disconnection reading collides with experimental facts (Bell-violation, non-separability) that any interpretation must accommodate.
- **Not that Leibniz or Malebranche are refuted.** Their systems are coherent; the point is that adopting their coherence is inheriting their empirical idleness.
- **No contested interpretation is promoted to fact.** The entanglement collision rests on V1 experimental results and the no-local-hidden-variables theorem; the separability-preserving escapes are flagged as the weakest sources (section 5.6) and shown to buy their rescue at the price of vacuity.
- **Not a rejection of the surviving core.** The structure-dependent-response lens is retained, stated as a prohibition, and recommended as the honest home of the thesis.

5.6 What the verdict rests on

Finally, the dependency map — what is robust and what is hostage to an open problem.

Load-bearing claim	Rests on	Status
Self-refutation of the strong reading	"a received signal is a relation"; structural coupling is a relation	conceptual, robust (V1)
Entanglement collision	Bell-violation; non-separability of entangled states	experimental, robust (V1), modulo interpretation
The pincer (vacuous-or-false)	occasionalist underdetermination	historical, robust (V1)
Separability-preserving escapes	superdeterminism / hidden-variable manoeuvres	contested, weakest (V3); shown idle
Necessary-only, open sufficiency	decomposition non-uniqueness	open problem, stated not solved
Surviving core (the lens)	structural determinism, constructivism	robust (V1/V2), as a prohibition

The argument leans only on the robust rows. The one genuinely open question — whether any decomposition-fixing rule *completely* closes the occasionalist re-description — is the thesis's analogue of the Putnam–Chalmers limit, and the paper states it rather than pretending to settle it.

A A worked demonstration: where the re-description is idle, and where it breaks

This appendix makes the two halves of the verdict concrete on a minimal model: the occasionalist re-description is *notation-only* in the separable regime, and *breaks* exactly at a joint statistic that no aggregate of orphans can reproduce. It is a *designed* illustration built on established results (the Bell–CHSH bounds), not a new measurement.

The setup. Two things, a and b, each presented with a local "signal" (a measurement setting): a with $x \in \{0, 1\}$, b with $y \in \{0, 1\}$. Each returns an outcome $A, B \in \{-1, +1\}$. A run is summarised by the correlation $E(x, y) = \langle A\, B \rangle$ and by the local marginals $\langle A \rangle, \langle B \rangle$.

Two models of the same data. The *relational* model lets the joint outcome depend on an irreducible relation between a and b. The *orphan* model is the disconnection thesis made explicit: each thing is a windowless bearer of intrinsic dispositions, so its outcome is a function of its own setting and a shared background variable λ drawn once,

$$A = f_a(x, \lambda), \qquad B = f_b(y, \lambda), \qquad \lambda \sim p(\lambda),$$

with no term coupling a's outcome to b's setting or vice versa. The λ is the "pre-coordination": the off-stage accommodation that lets two orphans' reactions agree without acting on one another — Leibniz's harmony, Malebranche's occasion, written as a hidden variable.

Where the re-description is idle. For *any* single fixed pair of settings, and for all one-sided marginals, the orphan model can match the relational model exactly: choose $p(\lambda)$ and the f's to reproduce the observed $\langle A \rangle$, $\langle B \rangle$, and $E(x, y)$. No single-setting measurement, and no marginal, distinguishes "the relation acted" from "the two orphans reacted, pre-coordinated by λ." This is the empirical equivalence of section 4.2 in miniature: the rewrite is total and lossless, and the relational fact has merely been relocated into $p(\lambda)$.

Where it breaks. The equivalence fails the instant we ask for a *joint* statistic across *several* settings. Form the CHSH combination

$$S \;=\; E(0,0) + E(0,1) + E(1,0) - E(1,1).$$

Every orphan model of the above form — every assignment of intrinsic dispositions plus a shared λ — satisfies the Bell–CHSH bound $|S| \leq 2$ (Clauser et al. 1969), the operational form of Bell's original inequality (Bell 1964). Entangled quantum systems reach $|S| = 2\sqrt{2} \approx 2.83$, and experiments measure values above 2 to high significance (Aspect et al. 1982). A measured $|S| > 2$ is therefore a joint fact that *no* aggregate of orphans can carry: it is a relational fact of the whole, irreducible to the parts' intrinsic dispositions and their arrangement.

What the demonstration shows. The same model delivers both halves of the paper's verdict. In the separable regime (single settings, marginals) the disconnection thesis is *unfalsifiable and idle* — it matches every number, having hidden the relation in λ. At the joint CHSH statistic it becomes *contentful and false* — it forbids $|S| > 2$, and nature crosses the bound. That is the pincer of section 3.1 on a single page: the disconnection thesis is exactly as testable as it is wrong, and exactly as safe as it is empty.

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